meadowlark optics

spatial light modulators

# User Manual

## Standard XY Phase Series

## Spatial Light Modulator

## With DVI Controller




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# 1 Introduction

This User Manual covers the XY Phase Series of liquid crystal on silicon spatial light modulators (SLMs) now available from Meadowlark Optics. The manual instructs users on how to set up the DVI electronic hardware, optics, and operate the system software. SLM system setup is fairly complex, incorporating optical components, mounting and positioning hardware, custom drive electronics hardware, and a proprietary software interface.

We strongly recommend that all users first read and familiarize themselves with the entire User Manual before initiating the setup and start up procedures.

## 1.1 Spatial Light Modulator Principles

An SLM is a device that modulates light according to a fixed spatial (pixel) pattern. The XY Phase Series SLMs convert digitized data into coherent optical information appropriate for a wide variety of applications, including beam steering, optical tweezers, diffractive optics, pulse shaping and more. These applications require modulators that can easily and rapidly change the wavefront of a coherent beam. By combining the electro-optical performance characteristics of liquid crystal materials with silicon-based digital circuitry, Meadowlark Optics now offers high speed SLMs that are also physically compact and optically efficient.

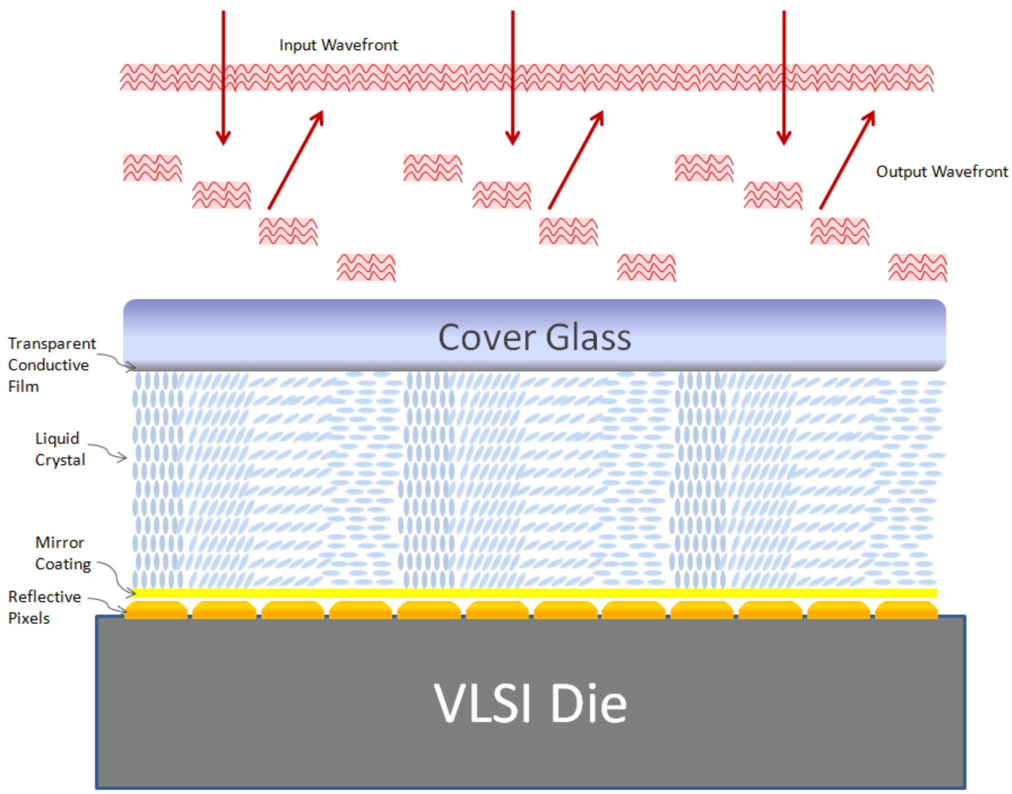


***Figure 1 ~ Cross sectional illustration of a liquid crystal spatial light modulator.***

Figure 1 shows the cross section of an LC SLM. Polarized light enters the device from the top, passes through the cover glass, transparent conductive film and liquid crystal layer, is reflected off the shiny pixel electrodes and returns on the same path. Drive signals from the driver boards travel through circuitry in the silicon backplane (VLSI die) to each pixel. The analog voltage induced on each electrode (pixel) produces an electric field between that pixel and the cover glass. This field produces a change in the optical properties of the LC layer. Because each pixel is independently controlled, a phase pattern may be generated by loading different voltages onto each pixel.

If requested by the customer, a dielectric mirror coating is applied to the backplane of the SLM. This coating is normally centered at a wavelength specified at the time of ordering. The dielectric mirror coating prevents diffraction from the inter-pixel regions of the backplane and increases the optical efficiency of the device.

## 1.2 Phase Modulation

The XY Phase Series of Spatial Light Modulators are fabricated with nematic liquid crystal, aligned in a homogenous configuration. Nematic liquid crystal has a variable electro-optic response to voltage. Figure 2 shows a simplified side view of a spatial light modulator's liquid crystal layer. When no voltage is applied to the LC, the molecules are parallel to the SLM coverglass and VLSI backplane. In this case, incident light will experience the largest difference between the extraordinary ($n_e$) and ordinary ($n_o$) index of refraction. As the voltage applied to the liquid crystal is increased, the LC will tilt until the extreme is reached and the LC is nearly normal to the SLM coverglass and VLSI backplane. At this point the difference between the extraordinary and ordinary index of refraction is nearly zero.

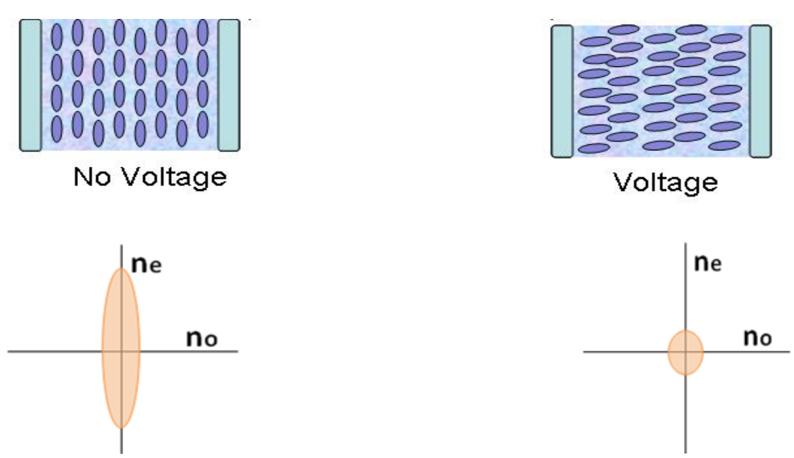


***Figure 2 ~ This diagram illustrates the liquid crystal orientation with respect to the coverglass and VLSI backplane as a function of applied voltage.***

***The molecules are parallel to the coverglass and backplane when no field is applied, and are nearly perpendicular to the coverglass and backplane when full field is applied.***

If light incident upon the SLM is linearly polarized parallel to the extraordinary axis, then a pure, voltage dependent phase shift will be observed. For example, if no voltage is applied to the pixel, the maximum phase retardation (typically a full wave at the design wavelength) will be applied. Likewise, if the pixel is programmed for maximum voltage, a minimum phase delay is applied. The result is a programmable phase on a per pixel basis.

Each of the SLM pixels is independently programmable to 65,535 discrete voltage states, all providing phase modulation. The response of the liquid crystal within the SLM to the applied pixel value (voltage) is nonlinear. To account for this, each SLM is shipped with a custom look-up-table (LUT). When this LUT is applied to an input image, the result is a linear output phase response ranging from 0 to $2\pi$.

## 1.3 Optical Test Setup

Depending on the application of the XY Phase Series SLM, many different optical setups can be used for either combined phase-amplitude mode or phase-only mode. Two examples of phase-only optical test setups are shown below.

### 1.3.1 Interferometer Setup

The first optical setup, illustrated in Figure 3, is a modification of a Twyman-Green interferometer (Handbook of Optics. Vol. 1, pp. 2.28-2.29). Here a monochromatic, collimated light source (i.e. laser beam expanded so it is larger than the diagonal of the SLM) passes through a non-polarizing beamsplitter such that the beam is divided into two beams, with nearly equal intensity. One of these beams illuminates the XY Phase Series SLM, while the other illuminates a reference mirror. Each of these reflected beams is then recombined at the image plane of a lens. If the reference mirror and the SLM are carefully aligned such that they are nearly coplanar, interference fringes will be visible at the image plane. A camera is placed at the image plane in order to magnify the fringes for easier viewing. When the XY Phase Series SLM is driven with different phase patterns, dynamic interference fringes can be viewed. Analyzing the interference fringes will then provide insight into the phase modulation provided by the XY Phase SLM.

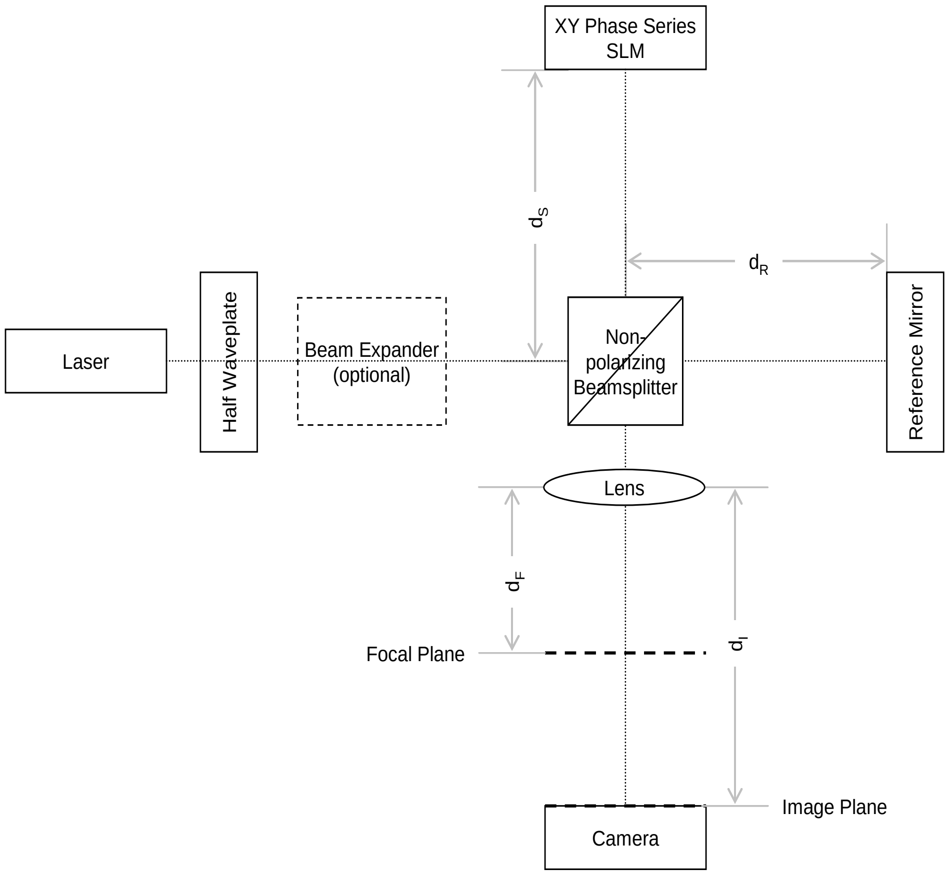


***Figure 3 ~ A Twyman-Green interferometer for testing a XY Phase or PhaseFlat Series SLM.***

In order to ease the alignment of the reference mirror and the XY Phase Series SLM in the Twyman-Green interferometer, place the camera or a card into the beam path at the focal plane of the lens. There should be two spots visible, one from the reference mirror and one from the SLM (there may also be very faint high-order diffraction spots from the SLM). If utilizing a beamsplitter cube, there are likely to be additional spots that are ghost images from the beamsplitter. When adjusting the tip/tilt of either the reference mirror or the SLM, the ghost spots will not move, but the true spots from the reference mirror and the SLM will move. Once the correct two spots are located, adjust the tip and tilt of the reference mirror and/or the SLM until these two spots are aligned on top of one another. Place the camera back in the image plane, or remove the card, and interference fringes should be visible in the image plane. By slowly adjusting the tip and/or tilt of just the reference mirror or SLM, the desired interference fringes should be readily obtained.

If working with a laser diode it is important to note that the coherence length of laser diodes are typically much shorter than the coherence length of a gas-discharge laser. With a Twyman-Green interferometer it is critical that both the SLM leg and the reference mirror leg have the same optical path length, ensuring that the short coherence length of a laser diode does not significantly reduce the visibility of the interference fringes. In order to ensure equal path lengths, set the camera such that it is focused onto the SLM – distinct features of the silicon backplane, such as individual pixels or the edge of the active area, should become sharp. Blocking the reference leg beam will likely be necessary in order to prevent crosstalk in the camera. Then move the block from the reference leg to the SLM leg and move the reference mirror closer to, or further from, the beam splitter until it is also in sharp focus on the camera. Since the reference mirror will likely not have any features with which to see the best focal point, look for focus of dust particles, or perhaps very carefully place a card onto the reference mirror such that it covers about half of the beam. The card edge can then be used as a guide to find the best focal position of the reference mirror.

There are a few important issues to remember whenever working with an SLM.

1. **Polarization** – The SLM is basically a variable single-order retardation plate, or wave plate. Like all wave plates, there is a fast axis and a slow axis. However with the XY Phase Series SLMs the index of refraction along the slow axis can be decreased electronically. When the light source is linearly polarized and parallel to the slow axis of the SLM, the result is phase-only modulation of the light source. If instead the light source were incorrectly oriented to be linearly polarized perpendicular to the slow axis, there would be no modulation observed with variable voltage. The use of a passive half-wave plate will greatly facilitate achieving the desired polarization alignment.

2. **Diffraction** – The SLM has discrete, reflective pixel pads in order to isolate the electrical signals and allow phase patterns to be written into the SLM. As a result of these discrete pixel pads, there will be diffraction. This diffraction can easily be seen in the Focal Plane of the Lens. There will be a very bright center spot ($0^{th}$-order), surrounded by a grid of spots becoming progressively dimmer as they get further from the $0^{th}$-order. In order to attain the maximum throughput, it is suggested to use as many of the diffracted spots, or orders, as possible. However, some applications do not allow the use of more than one order (typically the $0^{th}$-order).

3. **Dispersion** – Liquid crystal wave plates are not very achromatic because the index of refraction varies as a function of wavelength. This dispersion means that a device designed to provide a $2\pi$ phase stroke at one wavelength, will provide less than $2\pi$ phase stroke at a longer wavelength, and more than $2\pi$ phase stroke at a shorter wavelength.

4. **Optical Quality** – Due to the very small pixel pitch of the SLM, it is important to use high quality optics. A single element lens will generally have too much spherical aberration to provide a good, sharp focus across the entire clear aperture of the SLM. As a result, it is recommended to always use at least a doublet lens. For the same reason, the longer the focal length of the lens, the better the resulting image at the image plane. In addition, the transmitted wavefront distortion of most beamsplitter cubes is typically $\sim\lambda/4$, contributing to unacceptable wavefront distortion at the image plane. The use of a beamsplitter plate in place of a beamsplitter cube could improve the wavefront distortion,

but using a beamsplitter plate requires the use of a compensating plate in the reference leg of the Twyman-Green interferometer.

An off-axis setup, shown in Figure 4, is designed to maximize throughput by eliminating the non-polarizing beam splitter. A laser beam is incident on the SLM at a slight angle, reflects off of the reflective pixel pads, and then is imaged with a lens onto a camera. Please note that since this optical setup is not an interferometer, the actual phase modulation will not be visible on the camera. This optical setup is only shown to illustrate the concept of an off-axis system. The setup should be modified to meet the exact application requirements. The off-axis angle should be kept to a minimum in order to reduce crosstalk effects due to the beam traveling through more than one pixel region. Minimizing the off-axis angle also keeps the phase stroke closer to the designed value.

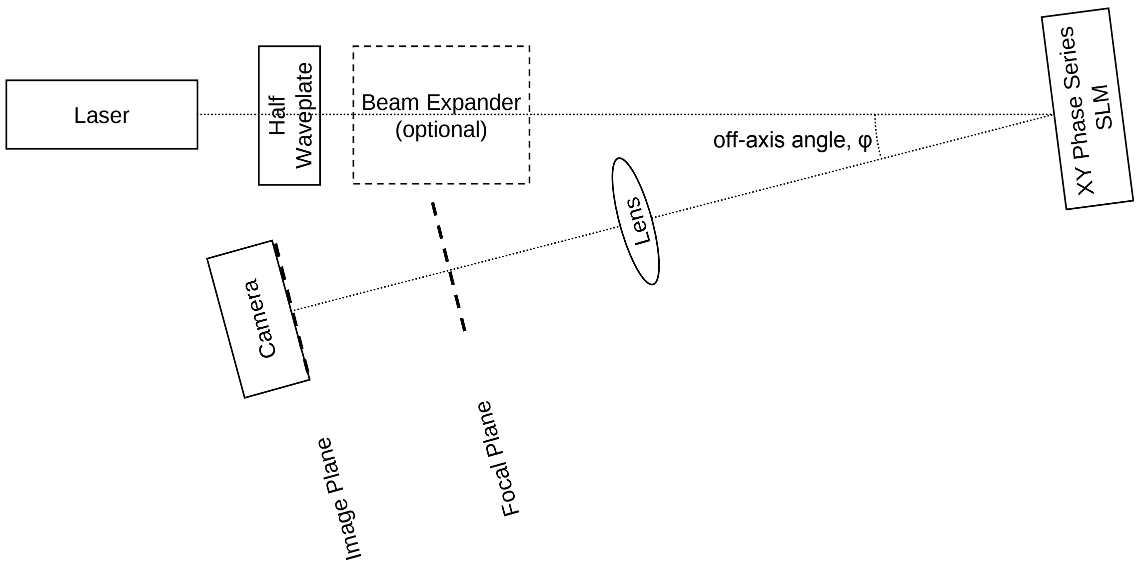


***Figure 4 ~ Off-axis optical setup for the XY Phase Series SLMs.***

## 2 Spatial Light Modulator System Shipping Contents

The complete 512 x 512 SLM system, which can be driven through a DVI interface, is shown in Figure 5. This system consists of a DVI Driver Box, the optical head containing the SLM and op-amp board, an external power cable and a DVI cable. Also included in the shipment and not shown in the image are a manual and a USB flash drive. The power cable must be connected from the DVI box to a power strip, the DVI cable must be connected from the DVI box to a graphics card in the computer, and the 80-pin ribbon cable must be connected from the DVI box to the op-amp board on the back of the SLM. Please note the white arrows on the cable to ensure proper keying when connecting.


***Figure 5 ~ Complete DVI system with all associated cabling.***

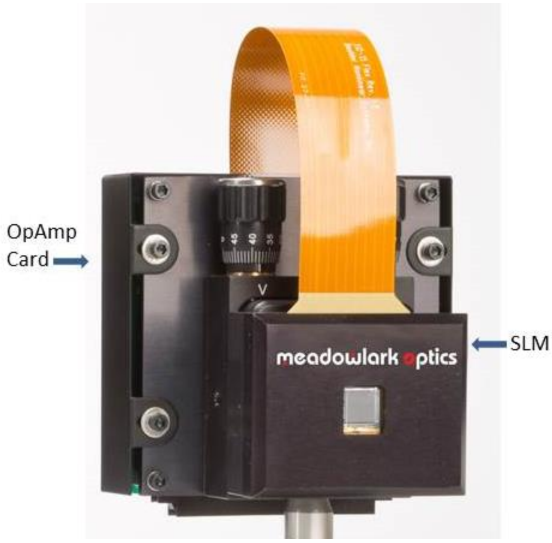


***Figure 6 ~ Optical Head Assembly, including the Op-Amp, and SLM.***

# 3 System Setup

**CAUTION**

**To minimize potential installation issues do not install any hardware until after the software has been fully installed.**

The diagram in Figure 7 shows the SLM system hardware. The analog data signals are then transferred to the op-amp board (on the SLM head assembly) where they are amplified to appropriate levels. A flex circuit is then used to transfer data from the op-amp board to the SLM itself.

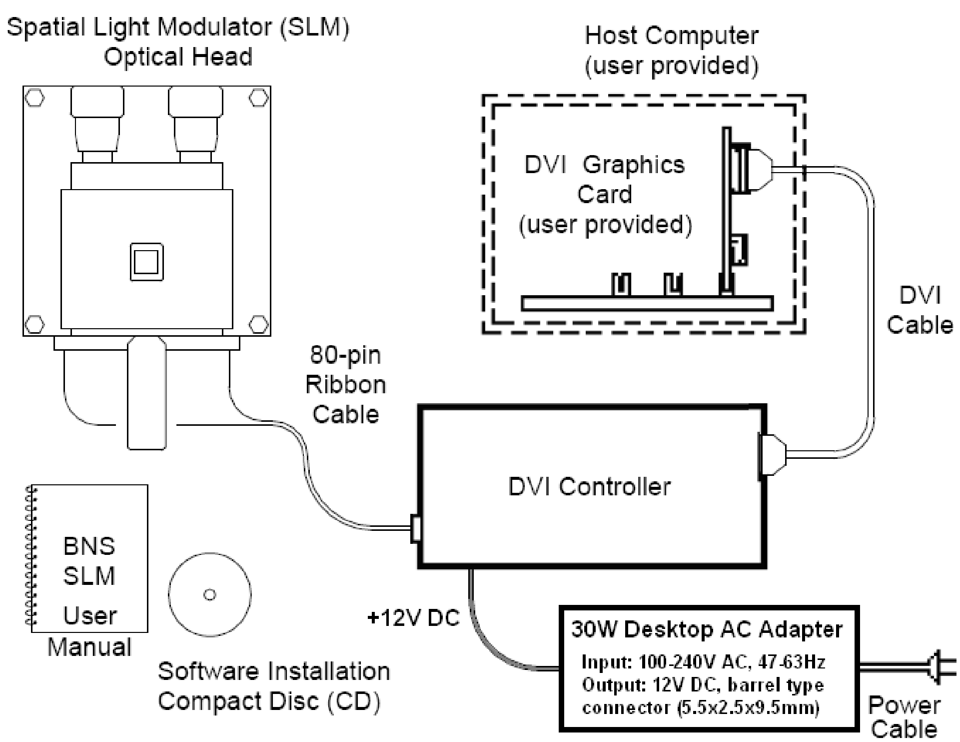


***Figure 7 ~ Top level diagram illustrating major components of the Meadowlark Optics SLM systems contents***

## 3.1 Software Installation

### *3.1.1 Minimum System Requirements*

In order to effectively utilize your SLM system some basic computing hardware is required. The following components are essential to properly achieve the full performance of your SLM system.

- IBM-compatible personal computer (PC), Pentium® -based (100 MHz minimum) system.

- USB port.
- Available DVI port.
- Windows®-based computer operating system (Windows 7 or 8).
- Mouse or other pointing device.
- Display monitor with 800 x 600 pixel format (minimum) with 24 or 32 bit color.
- 32 megabytes (MB) of available hard disk space is required for the Meadowlark Blink DVI basic software installation.
- 32 MB of available random access memory (RAM) to store and manage user-selected frames.

### *3.1.2 DVI Software Installation Instructions*

The Meadowlark DVI software on the USB flash drive contains the executable code necessary to operate the SLM, as well as sample pattern files, and various other support files. This software is used to load images to the SLM via the DVI interface. Please follow these steps to install the software.

1. Insert the USB that came with your SLM system into the USB drive.
2. Double click the setup installation file
3. When the Meadowlark DVI Install shield window appears, follow the onscreen instructions to complete the installation.
4. When the installation is complete, eject the USB drive from the drive and store it away carefully.
5. After installation is complete, the software should be ready to use. A shortcut to the executable will be installed on the desktop, and in the Start Menu.

## 3.2 Graphics Card Setup – 60 Hz System

Meadowlark has tested the 60 Hz system using a Nvidia Quadro NVS 290 with a DVI splitter connecting the graphics card output to both the primary monitor and the Meadowlark DVI system. When the system is initially connected it may be necessary to enable the SLM via the Windows Properties Dialog prior to being able to drive to the SLM. To do this, open the display properties window, right click on the secondary monitor, and select “Attached”.

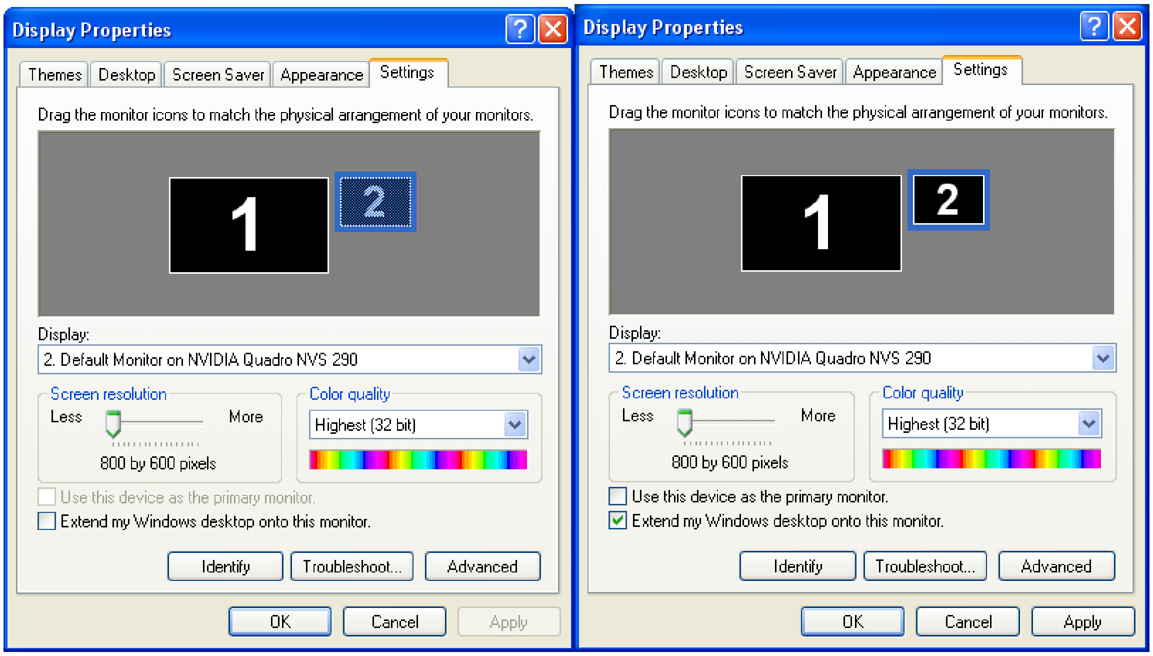


***Figure 8 ~Display Properties (a) prior to enabling the secondary "monitor" (which is the SLM), and (b) after enabling the secondary "monitor"***

Note the monitor arrangement with the primary monitor on the left, and the secondary monitor to the right. Color quality must be set to 32 bit in order for the device to work properly. If this arrangement is incorrect simply click and drag the secondary monitor to the desired location.

If using an Nvidia graphics card, open the Nvidia control panel, and set the display mode to 'Dualview'.

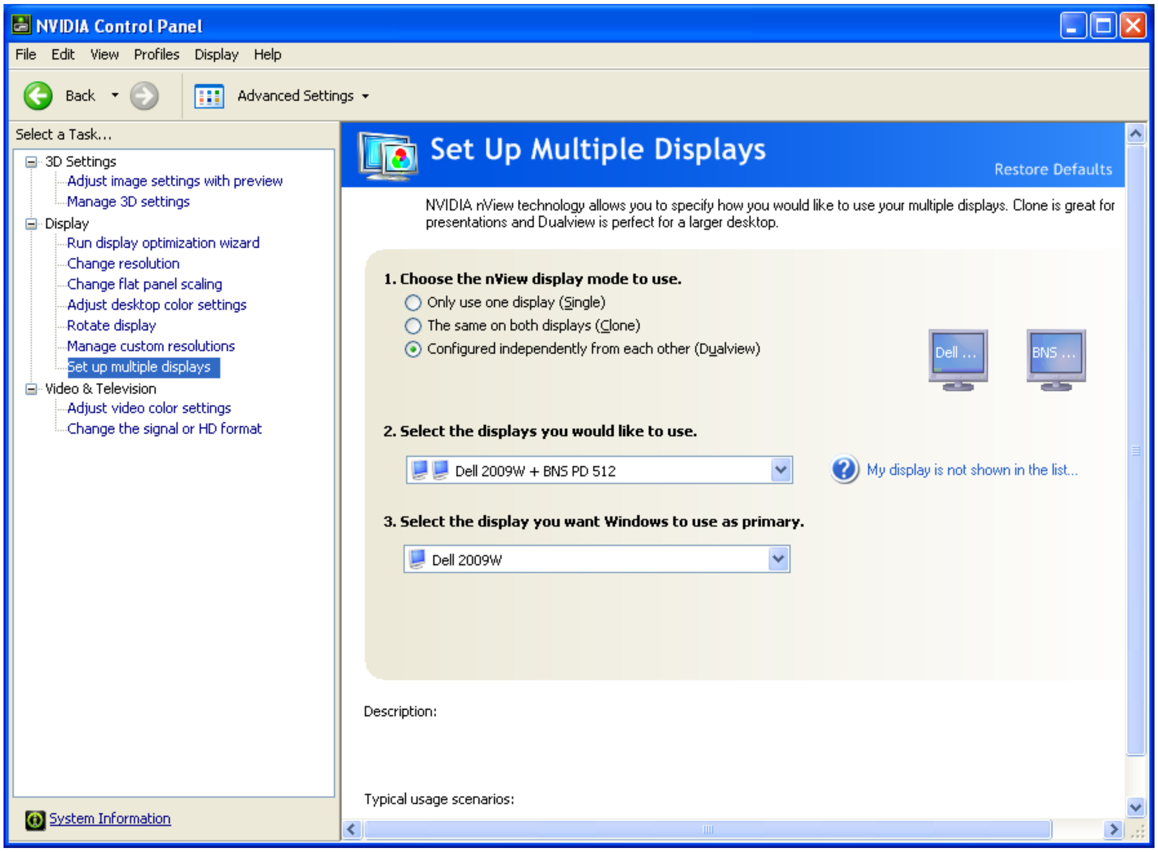


*Figure 9 ~ Dualview Setup in the Nvidia control panel*

At this point, the graphics card is properly configured to run in a 60 Hz configuration. Proceed to Section 4 for a description of the software operation.

## 3.3 Graphics Card Setup – 203 Hz System

Meadowlark recommends using an Nvidia Quadro GPU to support 200 Hz operation. A GPU with DVI Dual Link is required and it may be necessary to dedicate the GPU to the SLM. Meadowlark has tested 200 Hz operation on several Nvidia Quadro GPUs. Newer, untested models may work, but following models have been tested and recommended:

- Nvidia Quadro FX 5600
- Nvidia Quadro FX 580
- Nvidia Quadro 2000
- Nvidia Quadro K2000
- Nvidia Quadro K2000D
- Nvidia Quadro K620

After installing the graphics card, it will be necessary to enable the SLM and set the refresh rate via the Windows Properties dialog prior to being able to drive to the SLM. To do this, open the display properties window, right click on the secondary monitor, and select "Attached". A properly enabled system is shown in Figure 13.

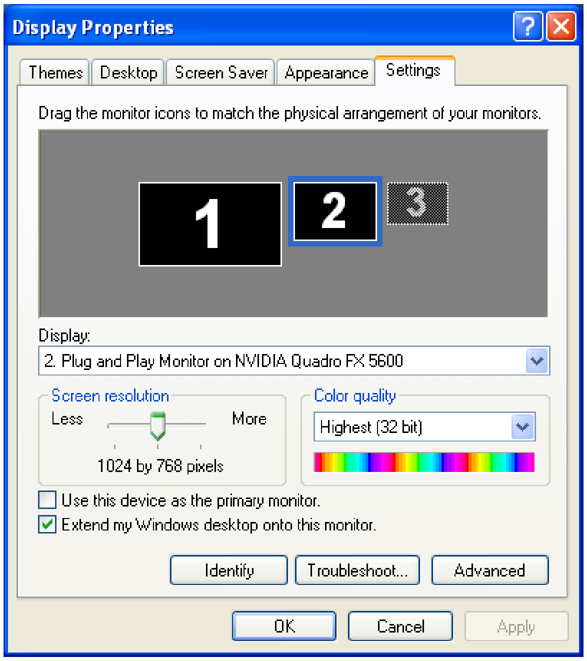


***Figure 10 ~ Multi Graphics Card Setup for 203 Hz system.***

***The primary is driven with a low end graphics card, and the SLM is driven with the Quadro FX 5600. One output of the 5600 is connected to the Meadowlark DVI hardware, while the other is left un-attached.***

Note in Figure 10 that the primary monitor is enabled, and the secondary monitor (the SLM) is enabled. The second output from the Quadro FX 5600 is not used, so the third monitor is disabled. Also note the order, with the primary monitor on the left, the secondary monitor in the middle, and the third unused monitor on the right. If this is not the order that the monitors show up in, then click and drag the monitors in the dialog of Figure 10 to the proper location. Color quality must be set to 32 bit in order for the device to work properly

After verifying that the SLM has been enabled, and is located to the right of the primary monitor, open the Nvidia control panel (if using an Nvidia graphics card) and verify the refresh rate. In the 'Change Resolution' dialog verify that the display resolution is set to 1024 x 768, and set the Refresh Rate to 203 Hz.

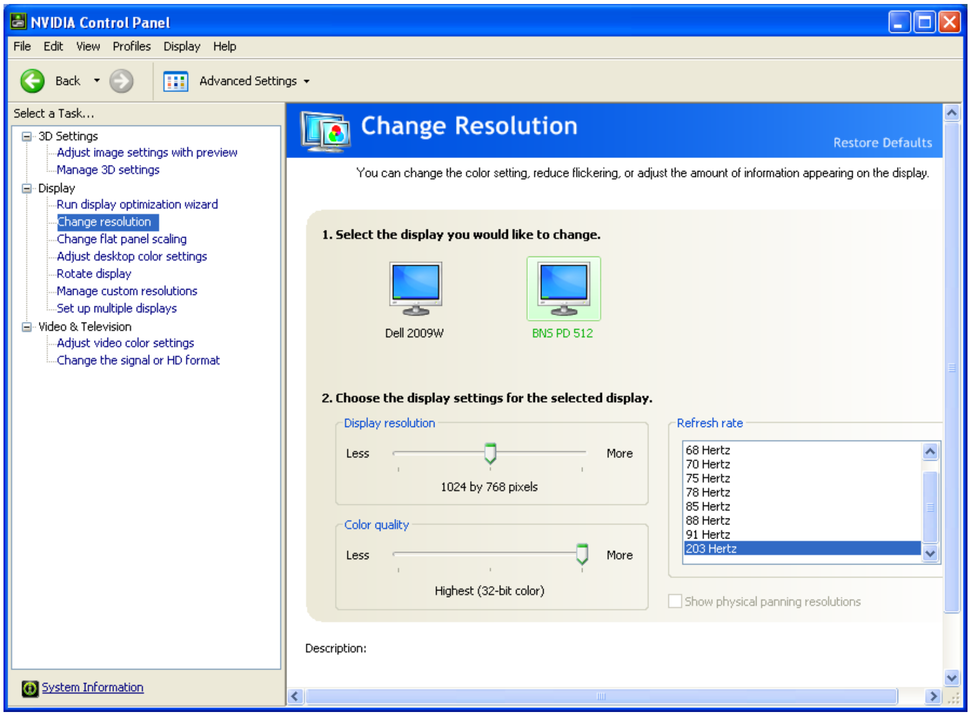


***Figure 11 ~ Resolution and Refresh Setup for the 203 Hz configuration.***

At this point, the graphics card is properly configured to run in a 203 Hz configuration. Proceed to Section 4 for a description of the software operation.

# 4 Software Operation

## 4.1 Meadowlark Blink DVI Operation

To start the software, double click the "Blink DVI" shortcut located on your desktop, or click on the Windows "Start" button, select, "Run…", type "C:\Blink\Blink_DVI.exe", then click "OK". When the software is started, the screen shown in Figure 12 should be visible.

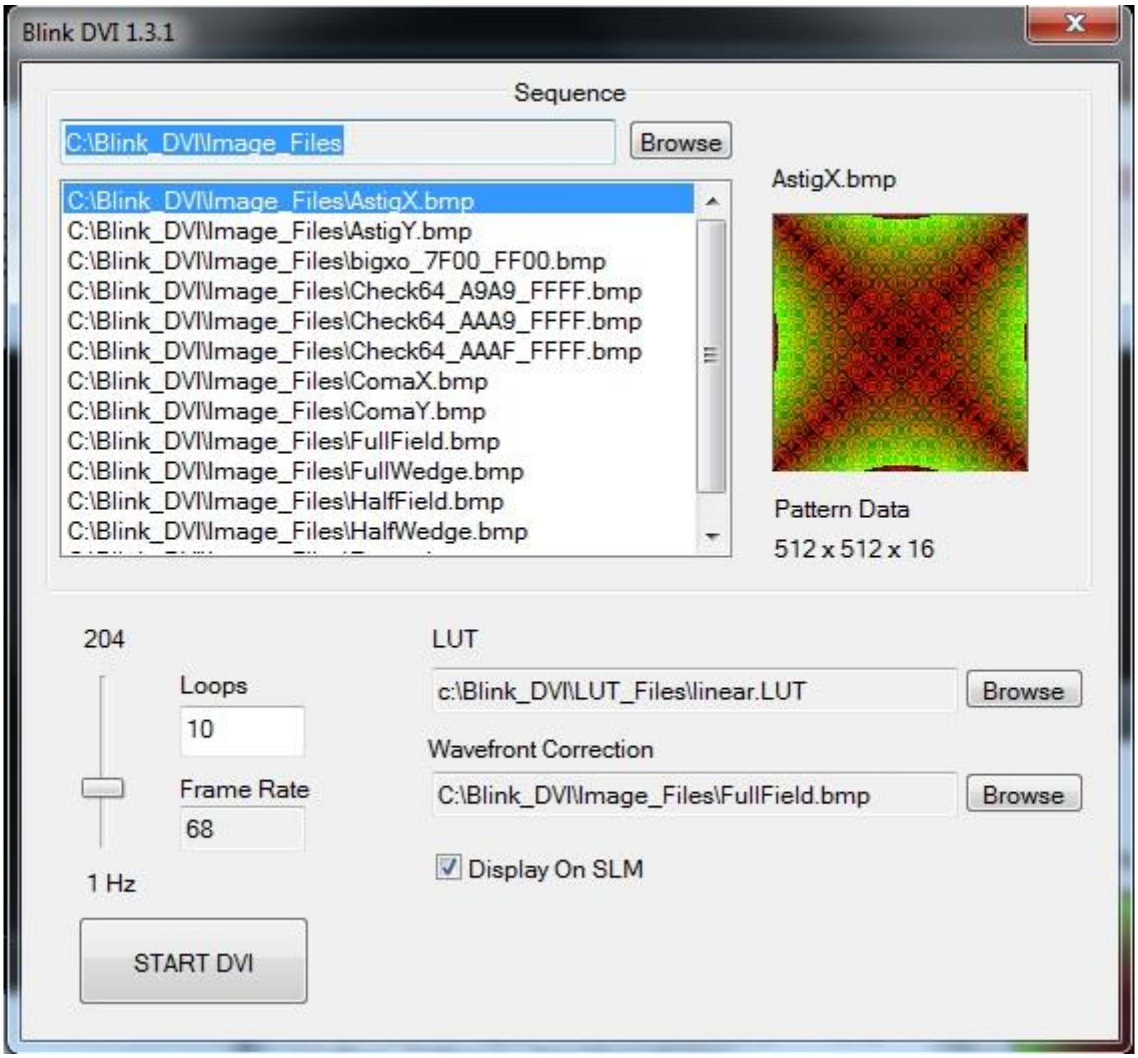


*Figure 12 ~ Main window for the Meadowlark Blink DVI software*

The SLM will be powered automatically when the power is applied to the DVI box. After the software has opened, the Pattern File selected in the Sequence List should appear both on the GUI and on the SLM if the 'Display On SLM' checkbox is checked. If the 'Display On SLM' checkbox remains unchecked, the image will appear in the upper left hand corner of the primary monitor. Toggling the state of this checkbox is a useful check to make sure the image data is being written correctly.

*Note: that the patterns in the GUI are in color. The DVI system is a 16 bit system. Typically 24 bit images are loaded to the hardware. In the case of a 24 bit image, the 8 green bits are the 8 most significant bits, the 8 red bits are the 8 least significant bits, and the 8 blue bits are ignored. In the case of 8 bit images, the 8 bits are fed directly into the 8 most significant bits of the hardware, and the image will appear green in the GUI. After reading the image, the raw image data is displayed on the GUI, and loaded directly to the hardware.*

## 4.2 Sequences

The Sequence list is defined by reading the contents of a folder and searching for bitmaps. The software stores information about the last folder used and loads the images from this folder upon initialization. Note that the images are not sorted, so to force a series of images to occur in a specific order it is recommended file names within a folder follow a numbering sequence such as: 000Img.bmp, 001Img.bmp, 002Img.bmp…etc.

To change the Pattern File currently loaded into the SLM, simply point the mouse at a different Pattern File in the Sequence List and click. This new Pattern File should now be highlighted, and be immediately loaded into the SLM. The cursor key (↑ and ↓) can also be used to select different Pattern Files in the Sequence list.

### *4.2.1 Running a sequence*

The Pattern Files can be changed repetitively by clicking the "START DVI" button located on the bottom of the main window. Once the "START DVI" button has been pressed, the system will sequence through all the Pattern Files in the current Sequence List a number of times that is set by the "Number of Loops" field. With the current value of 200, the list will be sequenced through 200 times and then stop.

The rate at which sequencing occurs is set by the Delay value. The Delay value can be set using the slider bar located just left of the "START DVI" button. The units for the Delay are in milliseconds (ms). This means that a certain number of milliseconds will pass before loading the next image in the sequence. Moving the slider bar will change the current value displayed in the Delay box.

### *4.2.2 Opening new sequence list*

To open a new sequence file move the cursor over the "Browse…" button and click. This opens the standard Windows wizard for opening a file or folder. The user is asked to locate a folder on the hard disk. Once located, the new sequence is automatically loaded into the system with the images currently contained within the selected folder. If the images within the folder are moved or deleted after loading into the system, it is necessary to click the "Browse…" button again and reselect the folder.

### *4.2.3 Look Up Table (LUT) Information*

Of primary interest to the user is the ability to change the LUT, or "look up table". A LUT is used to map the original input data to a set of new output values when the Pattern Files are loaded into the hardware. The LUT provides greater control by allowing the user to quickly change the response of a large set of Pattern Files without having to modify the Pattern Files themselves. Typically this is used to provide an approximately linear phase response to a linear set of pixel values. The LUT file is a text file with two columns of numbers. The first column is the input pixel value of your Pattern File. The second column contains the corresponding output pixel values that are loaded into the SLM hardware when a particular input pixel value is encountered in the Pattern File.

Each SLM system is shipped with a custom LUT designed to enable a linear phase stroke versus pixel value. This LUT will be named “slmXXXX.lut”, where XXXX corresponds to the serial number of your SLM. When using this LUT, a linear progression of input 0 – 65535 data should result in a linear output phase of 0 - 2π.

# 5 Mechanical Housing

The SLM Housing has vertical and horizontal micrometers for adjusting the tip and tilt up to ± 3°. The housing also allows the user to rotate the SLM about the optical axis by ~15°. A set screw (located beneath the flex, between the knobs) locks the rotation about the optical axis.

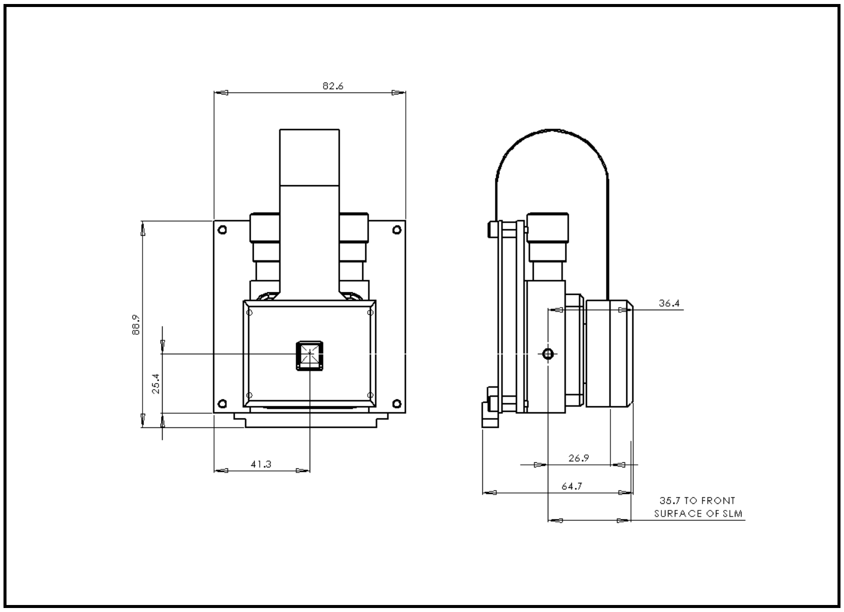


***Figure 13 ~ Outline drawing showing front and side views of 512 SLM Optical Head.***
***Dimensions in millimeters.***

# 6 SLM care and cleaning

**CAUTION**

**Failure to follow the recommendations included in this document may violate your warranty. Always ensure all personnel involved in the handling of your SLMs have appropriate tools, equipment, and training prior to cleaning any SLM.**

**Never use acetone to clean your SLM. Acetone will cause irreparable damage to your SLM.**

**Never use pressurized air or nitrogen to clean your SLM. Pressurized air or nitrogen will destroy the bond wires that provide the electrical interface to the SLM.**

The SLM is shipped with a piece of black paperboard covering the SLM aperture. This aperture, or a similar aperture, should be kept in place while the SLM is not is use to reduce the buildup of dust and debris on the SLM coverglass. When removing the aperture cover, or while the SLM coverglass is exposed, care should be taken to avoid any contact with the coverglass.

If dust or debris does contaminate the coverglass, gently remove them with a methanol soaked lens tissue. Take a small lens tissue or cleanroom wipe and soak a corner of it with methanol. Let the methanol evaporate until the lens tissue is just damp then gently wipe off the particles with the corner of the wipe. For more stubborn debris the methanol drag can be used. Fold a lens tissue several times until it is just small enough to fit inside of the SLM aperture. Soak the folded edge with methanol and let it evaporate until just damp. Line up the folded edge of the lens tissue with the edge of the glass, and drag the tissue straight across the glass. Never wipe the glass more than once with the same lens tissue as it will merely transfer dirt back onto the coverglass. Repeat as necessary.

If there is any dirt or contamination that cannot be removed, please contact the factory for additional cleaning instructions.

# 7 Troubleshooting

## 7.1 Image appears incorrect

When loading images to the SLM, make sure the image files are 512 x 512 24-bit bitmap images. If an image is smaller than 512 x 512, the SLM will display a bar across portions of the display containing the image and the other areas of the display will contain the last loaded image. If an image larger 512 x 512 is loaded, the software will no longer function and the program will have to be restarted.

## 7.2 When the computer boots the SLM is the primary monitor

Shut down, and disconnect the Meadowlark DVI system from the graphics card. Then boot without the SLM attached. The primary should now appear correctly.

# 8 System Specifications

## *Standard 512x512 SLM*

| | Model P512 – 0532 | Model P512 – 0635 | Model P512 – 0785 | Model P512 – 1064 | Model P512 – 1550 |
|---|---|---|---|---|---|
| **Array Size** | 7.68 x 7.68 mm | | | | |
| **Zero-Order Diffraction Efficiency** (standard) | 61.5% (maximum) | | | | |
| **Zero-Order Diffraction Efficiency** (with High Efficiency Mirror) | 90 - 95% (maximum) | | | | |
| **Duty Cycle** | Up to 100% | | | | |
| **External Window** - 600 to 1300 nm also available (see chart on page 12) | Broadband antireflection coated for $R_{avg}$ < 1% (over 450 - 865 nm) | | | Broadband antireflection coated for $R_{avg}$ < 1% (over 850 - 1650 nm) | |
| **Fill Factor** (standard product) | 83.4% | | | | |
| **Fill Factor** (with High Efficiency Mirror) | 100% | | | | |
| **Format** | 512 x 512 (262,144 active pixels) | | | | |
| **Mode** | Reflective | | | | |
| **Modulation** | Controllable index of refraction | | | | |
| **Phase Levels** (resolvable) | 50 linear levels (minimum) for 2π phase stroke with PCI or PCIe 8-bit Controller | | | 100 linear levels (minimum) for 2π phase stroke with PCI or PCIe 8-bit Controller | |
| | 1,000 – 8,000 linear levels (minimum) with DVI 16-bit Controller | | | 2,000 – 16,000 linear levels (minimum) for 2π phase stroke with DVI 16-bit Controller | |
| **Phase Stroke** (double-pass) | Typically 2π (π to 12π upon request) | | | | |
| **Contrast Ratio** (if used in Amplitude Mode) | 200:1 | | | | |
| **Pixel Pitch** | 15 x 15 μm | | | | |
| **Spatial Resolution** | 33 lp/mm | | | | |
| **Reflected Wavefront Distortion - RMS** (standard) | λ/3 @ 532 nm | λ/4 @ 635 nm | λ/5 @ 785 nm | λ/6 @ 1064 nm | λ/8 @ 1550 nm |
| **Reflected Wavefront Distortion – RMS** (PhaseFlat) | λ/12 @ 532 nm | λ/15 @ 635 nm | λ/20 @ 785 nm | λ/20 @ 1064 nm | λ/20 @ 1550 nm |
| **Standard Liquid Crystal** Response Time / Switching Frequency | ≤ 33.3 ms / ≥ 30 Hz | ≤ 33.3 ms / ≥ 30 Hz | ≤ 55.5 ms / ≥ 18 Hz | ≤ 66.7 ms / ≥ 15 Hz | ≤ 100 ms / ≥ 10 Hz |
| **High Speed Liquid Crystal** Response Time / Switching Frequency | ≤ 7 ms / ≥ 142 Hz | ≤ 12 ms / ≥ 83 Hz | ≤ 17.2 ms / ≥ 58 Hz | ≤ 10 ms / ≥ 100 Hz | ≤ 20 ms / ≥ 50 Hz |
| **High Efficiency with High Speed Liquid Crystal** Response Time / Switching Frequency | ≤ 10 ms / ≥ 100 Hz | ≤ 16.7 ms / ≥ 60 Hz | ≤ 22.2 ms / ≥ 45 Hz | ≤ 16.7 ms / ≥ 60 Hz | ≤ 28.5 ms / ≥ 35 Hz |
| **Wavelength Range** | 515 – 585 nm | 615 – 700 nm | 760 - 865 nm | 1030 - 1170 nm | 1505 - 1650 nm |

Above specifications are subject to change without notice. Please contact Meadowlark Optics for additional updates, and/or wavelength-specific data.

**Thank you** for purchasing a Meadowlark Optics Spatial Light Modulator.

For additional product and company information, please contact:

*Meadowlark Optics, Inc.*
*5964 Iris Parkway*
*P.O. Box 1000*
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